

DATA 557: Applied Statistics and Experimental Design

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Case Study Analysis: Faculty Salary Discrimination Analysis

1. Introduction

Disparities in compensation between male and female employees have been extensively recorded across various labor markets, including institutions of higher education in the United States. Despite the fact that salary discrimination based on gender is prohibited by law in the U.S., there continues to be a vigorous discussion regarding whether the salary differences observed between male and female faculty are indicative of discrimination or if they can be attributed to alternative factors.

Numerous elements may affect faculty compensation, such as academic discipline, years of experience, level of education, rank, administrative duties, and overall productivity. For instance, salary variations may arise across different fields due to market demand, and faculty members with greater experience or elevated academic ranks generally receive higher salaries. Furthermore, serving as a department chair or in other administrative capacities often involves additional compensation.

In this case study, we adopt a group-level analytical approach by analyzing salary trends among faculty members to determine if there are systematic differences in compensation between male and female faculty, while also taking into account other pertinent characteristics.

2. Data Description

The dataset comprises details regarding faculty salaries at a specific university in the United States. Data was collected for 1,597 faculty members who were employed from 1976 to 1995, with the exception of those in the medical school. Monthly salaries were documented for each faculty member across the years, culminating in a total of 19,792 observations. Each entry reflects the salary and relevant attributes of a faculty member for a particular year.

The dataset contains variables describing faculty demographics, education, employment history, academic field, rank, administrative responsibilities, and salary. A full description of all variables included in the dataset is provided in Table 1.

Table 1. *Description of variables in the dataset*

Variable	Description
case	Case number
id	Identification number for the faculty member
sex	M (male) or F (female)
deg	Highest degree attained: PhD, Prof (professional degree, e.g., medicine or law), or Other (Master's or Bachelor's degree)
yrdeg	Year highest degree attained
field	Arts (Arts and Humanities), Prof (professional school, i.e., Business, Law, Engineering, or Public Affairs), or Other
startyr	Year in which the faculty member was hired (2 digits)

year	Year (2 digits)
rank	Rank of the faculty member in this year: Assist (Assistant), Assoc (Associate), or Full (Full)
admin	Indicator of whether the faculty member had administrative duties (e.g., department chair) in this year: 1 (yes), or 0 (no)
salary	Monthly salary of the faculty member in this year, in dollars

3. Analysis

3.1 Question 1: Does wage discrimination exist in the starting salaries of faculty members (i.e., salaries in the year hired)?

Hypotheses:

To formally evaluate whether starting salaries differ by gender, we tested the following hypotheses:

- $H_0: \mu_{\text{male}} = \mu_{\text{female}}$ (the mean starting salary is the same for male and female faculty)
- $H_a: \mu_{\text{male}} \neq \mu_{\text{female}}$ (the mean starting salaries differ by gender).

Methods:

To examine whether wage discrimination exists in starting salaries, the dataset was first restricted to observations where the calendar year equals the hiring year (year = startyr). Restricting the dataset to the hiring year ensures that each faculty member contributes only one observation, allowing the salary comparisons to be treated as independent observations across individuals. This ensures that the salary variable represents the faculty member's starting salary at the time of hire.

Three analytical approaches were used:

1. A Welch two-sample t-test comparing the average starting salaries of male and female faculty.
2. A year-by-year comparison of starting salaries by gender.
3. A linear regression model controlling for rank, degree, and year of hire to evaluate whether gender remains a significant predictor of starting salary.

Visualizations using boxplots were also created to illustrate salary distributions by gender and by hiring year.

Results:

The starting salary for male faculty is on average higher than that for female faculty. The average starting salary for female faculty is \$3,600.32, while that for male faculty is \$4,015.55. The difference of approximately \$415 between the two is statistically significant ($p < 0.001$) based on Welch's two-sample t-test.

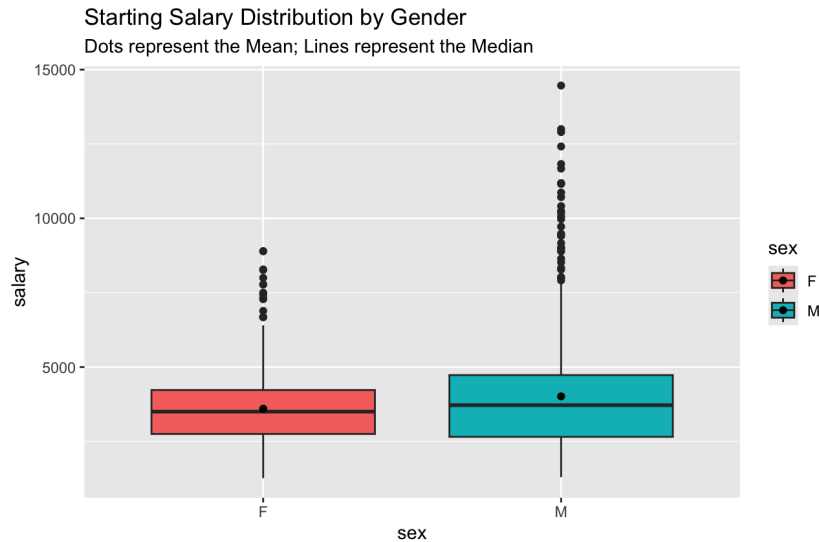


Figure 1. Starting Salary Distribution by Gender

Figure 1 shows the distribution of starting salaries by gender. The box plots for male teaching staff are usually higher than those for female teaching staff, and the average salary and median salary of males are both higher than those of females.

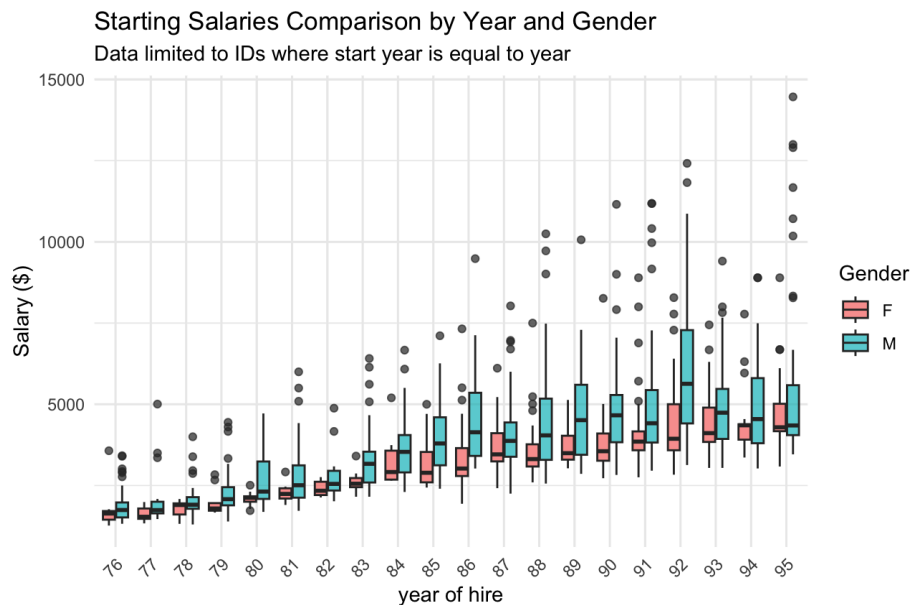


Figure 2. Starting Salaries by Year of Hire and Gender

From the perspective of salaries in different recruitment years, the salaries of male teaching staff are generally higher than those of female teaching staff in most years. Some differences between these years are statistically significant. Figure 2 shows this pattern, where the salary distribution of male teaching staff is usually higher than that of female teaching staff.

The results of a regression model that takes into account job level, educational background, and year of entry are also similar. The coefficient for male teaching staff is approximately \$301.75. This represents the expected difference in starting salary between male and female faculty with the same rank, degree, and year of hire. This difference is statistically significant.

Interpretation:

These results provide strong evidence of the existence of a gender pay gap within the teaching profession. Generally speaking, the starting salary of male faculty is typically higher than that of female faculty. It is worth noting that even after taking into account factors such as academic position level, educational background, and year of entry, this difference remains statistically significant.

These research findings indicate that the observed salary disparity cannot be entirely attributed to differences in educational qualifications or variations in recruitment conditions. Therefore, these results are consistent with the phenomenon of gender-based salary differences at the time of university admission.

Permutation Test:

To further validate these results and avoid relying on parameter assumptions, we conducted 10,000 iterations of a "permutation test". By randomly shuffling the gender labels (while keeping the salary data unchanged), we simulated the distribution of salary differences under the null hypothesis (i.e., gender has no effect on salary).

- The observed average difference value: 415.2
- The P value of the permutation test: 0.0002

The permutation distribution plot (as shown in the figure 3) indicates that the observed difference of 415.22 dollars under random conditions is extremely rare.

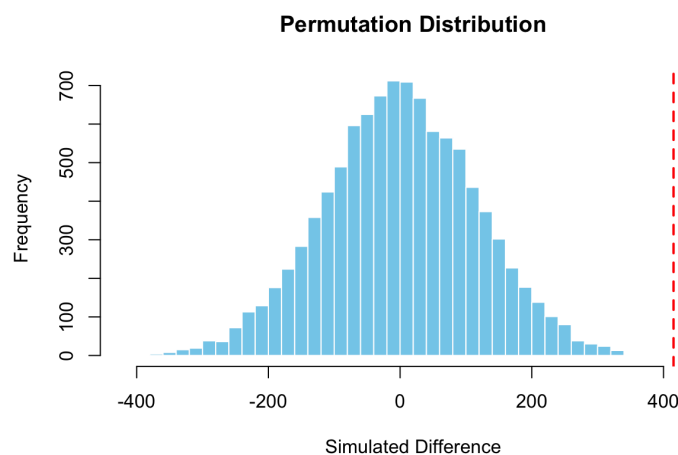


Figure 3. Permutation test results

Out of 10,000 trials, only 2 resulted in outcomes that were of the same degree of discrepancy as those observed in our dataset. This non-parametric evidence strongly supports the conclusion of the Welch t-test, confirming that the starting salary gap is highly unlikely to be caused by random events or outliers.

3.2 Question 2: Overall, how would you answer the question: Is there wage discrimination in salaries at the university?

Methods:

In order to assess the presence of wage discrimination within university faculty, we conducted an analysis of the most recent salary data for each faculty member, which reflects the cumulative impacts of hiring practices, promotions, and career progression. The predictor variables included sex (primary variable of interest), rank (Assistant, Associate, Full), highest degree attained, years of experience, administrative duties, and starting salary to control for initial gender differences at hire.

To assess whether the gender gap differs across academic ranks, we included an interaction term between sex and rank, treating rank as an effect modifier. Categorical variables were coded as factors in R, using Assistant Professors and female faculty as the reference groups. Observations with negative or zero years of experience were excluded to ensure meaningful interpretation.

Exploratory analyses included descriptive summaries of mean and median salaries by sex and rank, boxplots to visualize salary distributions across ranks, and a scatterplot of salary versus experience with separate trend lines for men and women to illustrate differences in salary growth trajectories (Figure 4).

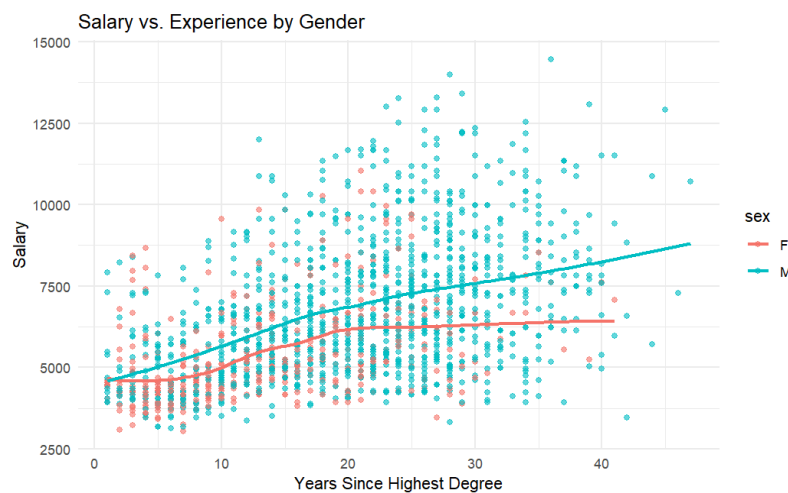


Figure 4. Scatterplot of salary vs. experience by gender

A multiple linear regression model was used to estimate the adjusted gender pay gap:

$$\text{Salary} = \beta_0 + \beta_1 \text{Sex} + \beta_2 \text{Rank} + \beta_3 \text{Degree} + \beta_4 \text{Experience} + \beta_5 \text{Admin} + \beta_6 \text{StartSalary} + \beta_7 (\text{Sex} \times \text{Rank}) + \epsilon$$

Hypotheses were tested as follows:

- $H_0 : \beta_1 = 0$ (no overall gender effect)
- $H_0 : \beta_7 = 0$ (gender gap does not vary by rank)

Results:

The mean salary for female faculty was \$5,397, compared to \$6,732 for male faculty, with median salaries of \$5,016 and \$6,313, respectively. Gender differences persisted across ranks (Figure 5).

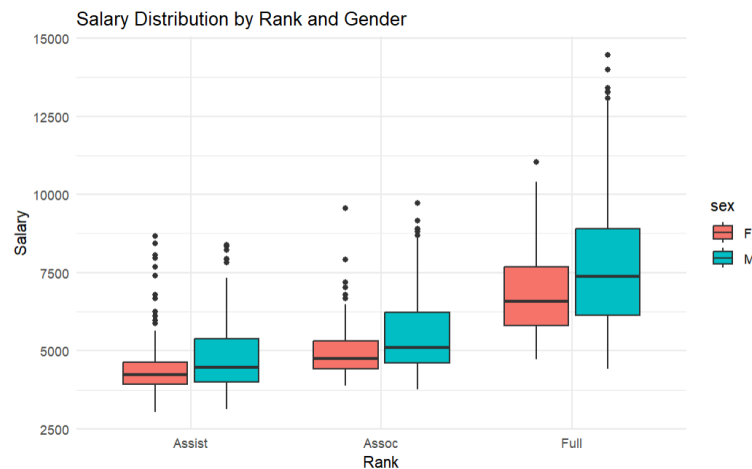


Figure 5. Boxplot of salary by sex and rank

The regression model explains 68.6% of the variation in salary and showed a minimal adjusted gender gap among Assistant Professors (approximately \$78, which was not statistically significant), which increased to about \$385 for Associate and \$435 for Full Professors, both borderline statistically significant ($p \approx 0.086$ and $p \approx 0.058$, respectively; Figure 6).

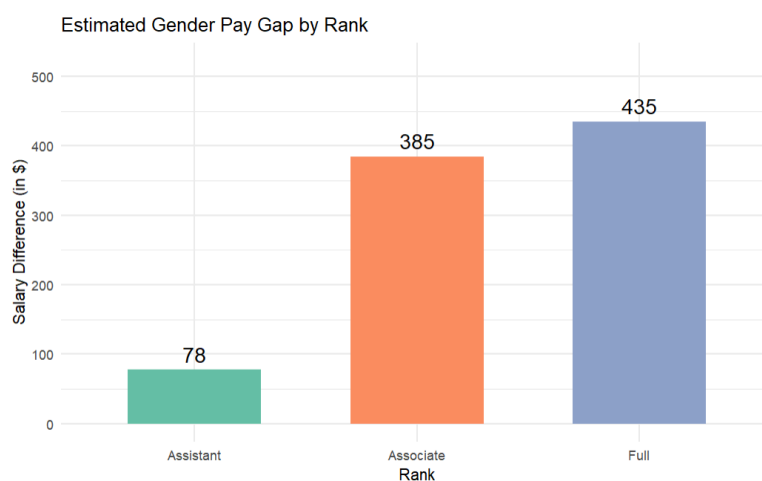


Figure 6. Adjusted Gender pay gap by rank

Other predictors accounted for a large portion of salary variation. Each year of experience contributed roughly \$34, administrative roles were associated with an additional \$987, and holding a PhD or professional degree increased salary by about \$500. Starting salary was the strongest predictor; each additional dollar at hire corresponded to approximately \$0.52 in current salary, underlining its lasting impact on earnings.

Interpretation:

This analysis shows that women earn less than men overall, largely due to lower starting salaries. The gender gap increases at higher ranks, although the differences are only borderline statistically significant after adjusting for experience, rank, administrative duties, and starting salary.

Starting salary acts as a confounder, explaining much of the observed overall wage gap, while rank functions as an effect modifier, with gender gaps increasing at higher academic levels. Experience, degree, and administrative roles account for much of the remaining variation in salaries.

3.3 Question 3: What issues are involved in generalizing your results?

The results of this analysis are based on data from a single U.S. university which places a significant constraint on how broadly the findings can be applied. Universities differ substantially in their size, mission, funding sources, and the mechanisms through which faculty salaries are determined. Some institutions rely heavily on external market benchmarks, particularly in fields like business, law, and engineering, while others apply more standardized internal pay scales tied to rank and seniority. The distribution of faculty across academic disciplines also varies widely across institutions. The proportion of faculty across fields, ranks, and degree types will differ across institutions. Because of this, gender pay gaps observed may be larger, smaller, or structured differently at other universities, and the findings should not be treated as representative of U.S. higher education broadly.

The data cover faculty employed between 1976 and 1995, a period during which the representation of women in academia, law enforcement, and institutional attitudes toward pay equity were all meaningfully different from today. The gender pay gap in starting salaries and the promotion disparities documented here reflect conditions from several decades ago. It is reasonable to expect that patterns have shifted since then, given increased awareness of pay equity, changes in federal enforcement of anti-discrimination law, and the growing share of women in faculty across various fields. Applying these findings to draw conclusions about wage discrimination at contemporary universities would require caution.

A related concern is that the dataset excludes medical school faculty entirely. Medical schools are often among the highest-paying units within a university and may have salary-setting practices and gender composition that differ substantially from the rest of the institution. The exclusion of this group means that the findings here do not represent all faculty at the university under study, let alone faculty more broadly. If salary discrimination patterns within the medical school differ from those observed in this analysis, the overall picture of pay equity at the institution would look different if those faculty were included.

Several variables that are known to influence faculty salaries are not available in this dataset. Productivity measures such as research grant funding, publication records, and teaching evaluations are commonly used in salary and promotion decisions but are not captured here. Teaching evaluations and service contributions, which also factor into performance reviews, are similarly absent. External market conditions matter as well, particularly in fields where universities compete with unmeasured dimensions, whether due to difference in career stage, field composition, or other structural factors, the estimated gender gaps in starting salary, promotion, and overall compensation may partly reflect those differences rather than discrimination alone. This residual confounding cannot be fully ruled out with the available data, and the gender effect estimates should be interpreted with that caveat in mind.

The structure and scope of the data also introduce methodological limitations that affect how results should be interpreted across the three analyses. In the promotion analysis, the fixed observation window ending in 1995 means that faculty who reached Associate rank in the later years of the study has very limited time to be promoted before the dataset ends. This right censoring problem is

particularly pronounced for the 1991 to 1995 cohort, where observed promotion rates are low not necessarily because promotion is less likely but because the follow-up period is too short to observe it. Similarly, some faculty who first appear in the dataset as Associate Professors in 1976 may have actually reached that rank before the observation window began, introducing left censoring in the time-to-promotion analysis. These censoring issues mean the promotion results for recent cohorts should be interpreted cautiously and would benefit from survival analysis methods that handle censored outcomes more formally. In the salary analysis for Question 3, using only the most recent salary observation per faculty member does not fully capture how the gender gap evolves over the course of a career. A longitudinal model that accounts for the repeated measures structure of the data and models salary growth trajectories would provide a richer and more precise picture of how compensation differences accumulate over time. Finally, the regression models in all three analyses assume linear relationships between continuous predictors and salary or promotion outcomes, which may not hold exactly in practice. Taken together, these methodological choices are reasonable given the structure and scope of the available data, but they represent constraints on the precision and depth of the conclusions that can be drawn.

4. Discussion and Limitations

This analysis applied a progression of statistical methods covered across the course to investigate gender-based wage discrimination among university faculty. The analysis began with exploratory analysis, using boxplots, summary statistics, and visualizations to identify initial patterns. These descriptive tools revealed male faculty consistently have higher salaries across starting pay, overall compensation, and promotion rates before formal testing was applied. Formal hypothesis testing was then used to evaluate whether the observed differences could be explained by chance. A Welch two sample t-test confirmed a statistically significant starting salary gap ($p < 0.001$), and a permutation test with 10,000 iterations validates this finding without relying on parametric assumptions. With only 2 out of 10,000 simulated differences as large as the observed \$415.22 gap, the non-parametric evidence strongly supported the t-test conclusion.

Linear regression was applied in both starting salary and overall salary analyses. In Question 1, after controlling for rank, degrees, and year of hire, male faculty still earned approximately \$302 more than female faculty at hire, confirming that the gap is not fully explained by differences in qualifications or hiring conditions. In Question 3, a multiple linear regression including an interaction term between gender and rank showed that rank functions as an effect modifier, with the adjusted gender gap growing from \$78 at the Assistant level to \$435 at Full Professor. Starting salary emerged discussed in the course: much of the overall salary gap traces directly back to the discrimination already present at the point of hire.

The promotion analysis in Question 2 applied logistic regression to model a binary promotion outcome. Before fitting the model, assumptions were verified: the outcome was binary, observations were independent, and variance inflation factors confirmed no problematic multicollinearity among predictors. The unadjusted odds ratio of 2.06 decreased to 1.71 after adjusting for years at Associate rank, field, degree, and cohort, consistent with the course concept of confounding. Residual diagnostics for the linear regression model of time to promotion, including a residuals versus fitted values plot and a Q-Q plot, confirm that the assumptions of homoscedasticity and approximate normality were reasonably satisfied.

Several limitations apply across all three analyses. The regression models assume linear relationships between predictors and outcomes, which may not exactly be true in practice. The overall salary analysis uses only each faculty member's most recent observation rather than modeling the full salary trajectory, and a longitudinal approach accounting for affected by right censoring, since faculty who reached Associate rank late in the study period had limited time to be promoted before the dataset ends in 1995. Handling this more formally would require survival analysis methods. Finally, key variables such as research productivity, grant funding, and publication output are absent from the dataset. Their absence introduces the possibility of residual confounding, meaning the estimated gender effects reflect meaningful associations but cannot be interpreted as fully causal estimates of discrimination.

5. Conclusion

5.1 Conclusion of the Analysis

Across all three questions examined in this study, the evidence points consistently towards the presence of gender-based pay disparities at this university. Starting salaries for male faculty were on average \$415 higher than for female faculty, a difference that held up under a t- test, a year-by-year comparison, a regression model controlling for rank, degree, and year of hire, and a non parametric permutation test. This convergence of methods provides strong and consistent evidence that the starting salary gap is real and not a product of chance or differences in qualifications at the time of hire.

In the promotion analysis, a male faculty were promoted from Associate to Full Professor at a higher observed rate (approximately 60% versus 42% for women). After adjusting for career characteristics, the odds of promotion for male faculty remained 71% higher than for female faculty, though this adjusted estimate fell just short of conventional statistical significance ($p=0.057$). Time to promotion and salary changes at promotion showed no meaningful differences between men and women, suggesting that once promotion does occur, the process is relatively equitable. The more pronounced disparity lies in who gets promoted and at what rate, rather than the mechanics of the promotion itself.

The overall salary analysis confirmed that women earn less than men at every rank. Much of this gap traces back to lower starting salaries, which compound over time given that starting salary is the strongest predictor of current salary. The gender gap also widens at higher academic ranks, which is consistent with the promotion disparities identified in Question 2: if women are less likely to reach Full Professor, they are also less likely to access the salary levels that come with that rank.

5.2 Conclusion of the Study

Taking the findings together, the evidence supports the conclusion that wage discrimination exists in this university, though it manifests differently across the faculty career. The clearest and most statistically robust evidence of discrimination is at the point of hire, where male faculty receive higher starting salaries than female faculty even after accounting for rank, degree, and year of hire. This initial disadvantage has lasting consequences because starting salary strongly predicts future earnings. The promotion analysis adds a further dimension: women face lower odds of being promoted to Full Professor, which limits their access to senior level compensation over the course of a career.

At the same time, the findings come with important qualifications. The data covers a single institution over a specific historical period, and the patterns observed here may not reflect the current state of

gender equity in academic compensation more broadly. Key variables such as research productivity and external market conditions are not available, meaning some of the estimated gender effects may partly reflect unmeasured differences between men and women rather than discrimination alone. The results should therefore be interpreted as evidence of systematic and statistically meaningful salary disparities that are consistent with discrimination, while acknowledging that the available data cannot definitively establish discrimination as the sole cause. Addressing these disparities fully would require not only more recent data and richer variable sets but also institutional commitment to regular pay equity review and transparent salary-setting practices.